

# Computational Analysis of Diesel–Biodiesel Blends in a Compression Ignition Engine Using ANSYS Forte

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**Abstract**—The depletion of fossil fuels and increasingly stringent emission regulations necessitate the adoption of alternative fuels for compression ignition (CI) engines. This study presents a comprehensive computational evaluation of dual-fuel combustion, focusing on biodiesel blends (B20, B50, and B100), alcohol–diesel mixtures, and the integration of nano-additives, situated within the broader body of published biodiesel–CFD literature. A computational fluid dynamics (CFD) approach using ANSYS Forte was employed to simulate a 4-stroke diesel engine operating at 1300 RPM, following a modeling philosophy consistent with prior sector-mesh biodiesel simulation studies [12], [14]. A 60° sector mesh with dynamic cell clustering was utilized to optimize computational efficiency while capturing complex auto-ignition and flame propagation phenomena [14], [15]. The simulation incorporated a custom multi-fuel chemical kinetics mechanism for a biodiesel composition consisting of n-heptane, methyl palmitate, and methyl oleate, in the spirit of the tri-component surrogate mechanisms established for biodiesel-fueled engine CFD [24], [26]. Results indicate that biodiesel blends yield lower carbon monoxide (CO) and unburnt hydrocarbon (UHC) emissions compared to pure diesel, with the simulated B100 case achieving an approximately 75% reduction in terminal UHC mass fraction relative to the diesel baseline – a trend consistent with experimentally reported UHC reductions of 21–28% for oxygenated and nano-additised biodiesel blends in the literature [1], [6]. This manuscript further synthesizes the physicochemical effects of nano-additives (e.g., carbon nanotubes, ZnO, Al<sub>2</sub>O<sub>3</sub>, CeO<sub>2</sub>) and bio-alcohols in resolving inherent biodiesel limitations, such as high viscosity and elevated NO<sub>x</sub> emissions [1], [3], [4], and introduces a mathematical framework – combining governing conservation equations, spray breakup kinetics, detailed chemistry solvers, and Linear-Programming/genetic-algorithm-based blend optimization – for systematically identifying blend ratios that balance combustion efficiency against the oxygen-driven NO<sub>x</sub> penalty. The findings provide a quantitatively grounded framework for optimizing renewable fuel blends to enhance brake thermal efficiency and mitigate tailpipe emissions.

**Index Terms**—ANSYS Forte, Biodiesel, Carbon Nanotubes, Compression Ignition Engine, Dual-Fuel Combustion, Extended Zeldovich Mechanism, Kelvin–Helmholtz/Rayleigh–Taylor Breakup, Linear Programming, Multi-Objective Optimization, Nano-additives, RNG  $k-\epsilon$  Turbulence Model, SAGE Chemistry Solver.

## I. INTRODUCTION

Global energy demands and the environmental impacts associated with fossil fuel consumption have driven extensive research toward sustainable alternative fuels for compression ignition (CI) engines [1], [2]. Biodiesel, an oxygen-rich monoalkyl ester derived from renewable feedstocks, has

emerged as a highly viable substitute capable of direct integration without significant engine modification [1]. Despite its environmental advantages, the widespread commercialization of pure biodiesel is hindered by adverse physicochemical properties, including high kinematic viscosity, inferior cold-flow characteristics, oxidative instability, and a tendency to increase nitrogen oxide (NO<sub>x</sub>) emissions during combustion [1], [4].

To mitigate these limitations and optimize engine performance, recent advancements have focused on blending strategies incorporating oxygenated alcohols and property-enhancing nano-additives [1], [8]. Blends of ethanol and diesel fuel have been shown to improve particulate matter (PM) reduction due to their oxygenated nature; experimental work has reported PM reductions of up to 68% with 50 vol.% n-butanol or 20 vol.% ethanol addition, though such blends typically require co-solvents or emulsifiers to maintain blend stability at lower temperatures and to restore lubricity [8]. Furthermore, the introduction of alcohol–diesel blends utilizing heavier alcohols, such as butanol or pentanol, correlates with a measurable reduction in the primary particle size of soot agglomerates [9], with diffusion-flame studies reporting soot-volume reductions from  $5.54 \times 10^{-4} \text{ mm}^3$  for neat diesel to as low as  $1.96 \times 10^{-4} \text{ mm}^3$  for a 50% n-butanol blend [10], favorably modifying in-cylinder soot-precursor formation and oxidation processes. Concurrently, nanotechnology offers substantial combustion improvements; nano-additives such as carbon nanotubes (CNTs), zinc oxide (ZnO), and alumina (Al<sub>2</sub>O<sub>3</sub>), when dispersed uniformly via ultrasonication, provide an increased surface area that enhances fuel atomization, accelerates oxidation, and improves overall brake thermal efficiency (BTE) while simultaneously lowering specific emissions [1], [2].

While experimental testing provides necessary empirical validation for these novel fuels, it remains highly resource-intensive. Computational Fluid Dynamics (CFD) software, specifically ANSYS Forte, serves as an essential analytical tool in evaluating complex combustion dynamics, successfully modeling the concurrent modes of auto-ignition and flame propagation characteristic of dual-fuel scenarios [14], [15]. By leveraging robust fuel chemistry models alongside targeted sector mesh generation techniques – such as utilizing a 60° sector to represent a symmetric cylinder and multi-hole injector configuration – computational burdens are signif-

icantly reduced without compromising simulation fidelity [14]. Sector-mesh methodologies of this kind have been validated in prior FORTE-based biodiesel studies, including simulations of waste-cooking-oil biodiesel (WB20) using methyl-palmitate and dodecane surrogates that closely reproduced in-cylinder pressure, temperature, and NO/CO mass-fraction contours obtained experimentally [12], and finite-volume Navier–Stokes/ $k-\epsilon$  half-cycle simulations of Jatropha biodiesel blends [13].

The primary objective of this study is to systematically compare the performance and emission characteristics of a diesel engine fueled by pure diesel against 20% (B20), 50% (B50), and 100% (B100) biodiesel blends. Through ANSYS Forte simulations, this research analyzes critical output parameters, including in-cylinder pressure, temperature distribution, and tailpipe emissions (CO, NO<sub>2</sub>, and unburnt hydrocarbons), and extends the analysis with (i) a synthesis of the published nano-additive and alcohol-oxygenate literature, (ii) the governing mathematical formulation underlying the CFD solver – covering conservation laws, turbulence closure, Lagrangian spray breakup, detailed chemical kinetics, and pollutant-formation kinetics – and (iii) a Linear-Programming and genetic-algorithm-based mathematical framework for optimal blend selection, mirroring optimization strategies already validated in the diesel/biodiesel/alcohol blending literature [27], [28], [30]. The study aims to establish blending configurations that maximize combustion efficiency while minimizing the environmental footprint of modern CI engines.

## II. RELATED WORK

### A. CFD Simulation of Biodiesel Combustion

Multiple studies have applied Reynolds-Averaged Navier–Stokes (RANS)-based CFD codes to biodiesel combustion. Prabhakaran and co-workers used ANSYS-Fluent with a non-premixed  $k-\epsilon$  RNG model and a moving combustion-chamber mesh to simulate mixture formation, combustion, and emissions of biodiesel–hydrogen blends [11]. Elumalai *et al.* employed ANSYS FORTE with methyl-palmitate and dodecane surrogates to simulate a naturally aspirated single-cylinder CI engine fueled by waste-coconut-oil biodiesel (WB20), reporting close agreement between simulated and experimental in-cylinder pressure, temperature, and species mass-fraction contours [12]. Kumar *et al.* performed half-cycle, finite-volume Navier–Stokes/ $k-\epsilon$  simulations of a single-cylinder DI CI engine operating on Jatropha biodiesel blends at 1500 rpm [13]. Ariffin and Alzurairi similarly investigated wall heat transfer using ANSYS Forte’s RNG  $k-\epsilon$  closure coupled with a 36-species/76-reaction skeletal n-heptane mechanism as a diesel surrogate [15]. These studies collectively establish sector-mesh, RNG  $k-\epsilon$ , detailed-chemistry CFD as the dominant methodology for biodiesel engine simulation, consistent with the approach adopted in the present work.

### B. Nano-Additive Effects on Biodiesel Combustion

A substantial body of experimental literature quantifies the performance and emission benefits of dispersing metal-oxide and carbon-based nanoparticles in biodiesel blends. Table I

TABLE I  
REPRESENTATIVE QUANTITATIVE EFFECTS OF NANO-ADDITIVES ON  
BODIESEL-FUELED CI ENGINE PERFORMANCE AND EMISSIONS, DRAWN  
FROM THE LITERATURE.

| Additive                                                | Reported Benefit                             | Source |
|---------------------------------------------------------|----------------------------------------------|--------|
| TiO <sub>2</sub>                                        | Smoke –32.98%, CO –30%, HC –28.68%           | [1]    |
| Al <sub>2</sub> O <sub>3</sub> (J20A1100)               | BTE +6.5%                                    | [2]    |
| CNT (J20C50)                                            | NO <sub>x</sub> –3.92%, CO reduction         | [2]    |
| MWCNT                                                   | BTE +36.81%                                  | [3]    |
| CeO <sub>2</sub>                                        | BSFC –30%                                    | [3]    |
| GNP                                                     | CO –65% (HC increase)                        | [3]    |
| CeO <sub>2</sub> +TiO <sub>2</sub> (150 ppm)            | BTE 28.9 → 31.5%, BSFC –7%, HC –21%, CO –16% | [4]    |
| Hybrid TiO <sub>2</sub> :Al <sub>2</sub> O <sub>3</sub> | BTE +12.5%, HC –20%, CO –16%                 | [5]    |
| Fe <sub>3</sub> O <sub>4</sub> (100 ppm)                | BTE +6.69%, CO –23.92%, HC –22.42%           | [6]    |
| Ni <sub>2</sub> O <sub>3</sub> (100 ppm)                | BTE → 24.89%, NO <sub>x</sub> –12 to –21%    | [7]    |

summarizes representative quantitative findings drawn from recent reviews and experimental studies. TiO<sub>2</sub> nanoparticles have been reported to reduce smoke opacity by 32.98%, CO by 30%, and HC by 28.68% [1]. Al<sub>2</sub>O<sub>3</sub>-doped Jatropha biodiesel (J20A1100) achieved a maximum 6.5% improvement in thermal efficiency, while CNT-doped blends (J20C50) produced the largest CO reductions and a 3.92% NO<sub>x</sub> reduction relative to the base blend [2]. Multi-walled carbon nanotubes (MWCNTs) have been shown to increase BTE by up to 36.81%, while CeO<sub>2</sub> reduces brake specific fuel consumption (BSFC) by as much as 30%, and graphene nanoplatelets (GNPs) reduce CO emissions by a minimum of 65% [3]. Combined CeO<sub>2</sub>/TiO<sub>2</sub> doping at 150 ppm raised BTE from 28.9% (diesel baseline) to 31.5% while lowering BSFC from 0.29 to 0.27 kg/kWh, with simultaneous reductions of 21% (HC), 16% (CO), and 24% (smoke opacity), at the cost of 9% and 7% increases in CO<sub>2</sub> and NO<sub>x</sub>, respectively [4]. Hybrid TiO<sub>2</sub>–Al<sub>2</sub>O<sub>3</sub> nanoparticle blends in B20 waste-cooking-oil biodiesel achieved thermal-efficiency gains of up to 12.5% and simultaneous HC and CO reductions of up to 20% and 16%, respectively [5]. Fe<sub>3</sub>O<sub>4</sub>-doped Sterculia foetida biodiesel blends increased BTE by 6.69% and reduced CO, HC, NO<sub>x</sub>, and smoke by 23.92%, 22.42%, 5.38%, and 3.61%, respectively [6]. Ni<sub>2</sub>O<sub>3</sub>-doped B20/B20But10 blends raised peak cylinder pressure to 55.86 bar and BTE to 24.89%, with HC, smoke, and NO<sub>x</sub> reductions of 13–28%, 8–43%, and 12–21%, respectively [7].

### C. Spray Breakup and Chemical Kinetics Modeling

Lagrangian–Eulerian spray modeling in modern engine CFD codes almost universally employs the Kelvin–Helmholtz/Rayleigh–Taylor (KH-RT) hybrid breakup model originally formulated by Reitz and Beale, which combines aerodynamically driven KH surface-wave instabilities governing primary breakup with RT instabilities governing secondary droplet breakup under strong deceleration [18]. Recent work has focused on recalibrating the KH-RT model constants

(particularly the breakup switching constant  $C_b$ ) as functions of ambient density and fuel physical properties, extending the model's applicability to biodiesel, DME, and methanol sprays [19]. Comparative assessments confirm that the modified KH-RT model provides the most physically consistent predictions among available breakup models for high-pressure dense diesel sprays [20]. On the chemistry side, detailed kinetic mechanisms for biodiesel surrogates have evolved from the two-component methyl-butanoate/n-heptane mechanism of Brakora *et al.* [24], [25] to tri-component skeletal mechanisms combining methyl decanoate, methyl-9-decenoate, and n-heptane (115 species, 460 reactions), which better capture the saturated and unsaturated fatty-acid-methyl-ester content of real biodiesel [26]. To manage the computational cost of coupling such detailed mechanisms with 3-D CFD, multi-zone acceleration algorithms – in which cells of similar thermochemical state are clustered into representative zones before stiff-ODE integration – are used in conjunction with detailed chemistry solvers such as SAGE [16], [17].

#### D. Blend and Operating-Parameter Optimization

Beyond CFD and experimental characterization, several studies have applied formal optimization frameworks to biodiesel blend selection. Hoang *et al.* combined Linear Programming (LP) with the Analytical Hierarchy Process (AHP) to select an optimal diesel/biodiesel/alcohol blend under simultaneous performance, emission, cost, and safety constraints, identifying a 70% diesel/20% biodiesel/10% butanol blend as optimal, with 35% lower sulphur content and 36% lower CO<sub>2</sub> emissions relative to baseline diesel [27]. Multi-objective evolutionary algorithms – including NSGA-II, NSGA-III, and AGE-MOEA – have been used to identify Pareto-optimal operating points balancing BTE, BSFC, and NOx across biodiesel/biokerosene blend ratios, compression ratios, and engine loads [28], [29]. Hybrid ANN-RSM frameworks combined with metaheuristics such as the Advantage-Critic Kronecker-factored Trust Region-Differential Evolution (ACKTR-DE) and Harris Hawks Optimization (HHO) algorithms have further been used to optimize waste-cooking-oil biodiesel/diesel blend ratios and injection parameters [30]. These optimization methodologies motivate the Linear Programming and genetic-algorithm formulations developed in Section V of this work.

### III. GOVERNING EQUATIONS AND COMPUTATIONAL FRAMEWORK

This section presents the mathematical formulation underlying the ANSYS Forte CFD solver, spanning the Reynolds-Averaged Navier–Stokes (RANS) conservation equations, RNG  $k$ - $\varepsilon$  turbulence closure, Lagrangian spray breakup kinetics, the Arrhenius-based multi-fuel chemistry solver, and the extended Zeldovich mechanism governing thermal NOx formation.

#### A. Conservation Equations

The gas-phase flow field is governed by the Favre (density-weighted) averaged compressible, reacting Navier–Stokes

equations. Continuity (with a Lagrangian spray mass source  $\dot{S}_M$ ):

$$\frac{\partial \bar{\rho}}{\partial t} + \frac{\partial (\bar{\rho} \tilde{u}_i)}{\partial x_i} = \dot{S}_M \quad (1)$$

Momentum:

$$\frac{\partial (\bar{\rho} \tilde{u}_i)}{\partial t} + \frac{\partial (\bar{\rho} \tilde{u}_i \tilde{u}_j)}{\partial x_j} = -\frac{\partial \bar{p}}{\partial x_i} + \frac{\partial \tau_{ij}}{\partial x_j} + \dot{S}_{M,i} \quad (2)$$

where the effective stress tensor is

$$\tau_{ij} = (\mu + \mu_t) \left( \frac{\partial \tilde{u}_i}{\partial x_j} + \frac{\partial \tilde{u}_j}{\partial x_i} - \frac{2}{3} \delta_{ij} \frac{\partial \tilde{u}_k}{\partial x_k} \right) \quad (3)$$

Energy conservation:

$$\frac{\partial (\bar{\rho} \tilde{e})}{\partial t} + \frac{\partial (\bar{\rho} \tilde{u}_j \tilde{e})}{\partial x_j} = -\bar{p} \frac{\partial \tilde{u}_j}{\partial x_j} + \frac{\partial}{\partial x_j} \left[ \left( \frac{\lambda}{c_p} + \frac{\mu_t}{Pr_t} \right) \frac{\partial \tilde{h}}{\partial x_j} \right] + \dot{Q}_{chem} + \dot{Q}_{spray} \quad (4)$$

Species transport, for  $k = 1, \dots, N_{sp}$  gas-phase species:

$$\frac{\partial (\bar{\rho} \tilde{Y}_k)}{\partial t} + \frac{\partial (\bar{\rho} \tilde{u}_j \tilde{Y}_k)}{\partial x_j} = \frac{\partial}{\partial x_j} \left[ \left( \rho D_k + \frac{\mu_t}{Sc_t} \right) \frac{\partial \tilde{Y}_k}{\partial x_j} \right] + \dot{\omega}_k + \dot{S}_{M,k} \quad (5)$$

where  $\dot{\omega}_k$  is the net chemical production rate of species  $k$  obtained from the kinetics solver (Section III-D), and  $\dot{Q}_{chem}$ ,  $\dot{S}_{M,k}$  couple the Eulerian gas phase to the Lagrangian spray and chemistry source terms, consistent with the coupling strategy of RANS-based diesel/biodiesel CFD codes [13], [15].

#### B. Turbulence Closure: RNG $k$ - $\varepsilon$ Model

In-cylinder turbulence was closed using the renormalization-group (RNG)  $k$ - $\varepsilon$  model, selected consistent with prior ANSYS Forte biodiesel-combustion studies [15]:

$$\begin{aligned} \frac{\partial (\rho k)}{\partial t} + \frac{\partial (\rho u_j k)}{\partial x_j} &= \frac{\partial}{\partial x_j} \left[ \alpha_k \mu_{eff} \frac{\partial k}{\partial x_j} \right] + G_k - \rho \varepsilon \quad (6) \\ \frac{\partial (\rho \varepsilon)}{\partial t} + \frac{\partial (\rho u_j \varepsilon)}{\partial x_j} &= \frac{\partial}{\partial x_j} \left[ \alpha_\varepsilon \mu_{eff} \frac{\partial \varepsilon}{\partial x_j} \right] + C_{1\varepsilon} \frac{\varepsilon}{k} G_k - C_{2\varepsilon}^* \rho \frac{\varepsilon^2}{k} \quad (7) \end{aligned}$$

with the RNG correction to the destruction term:

$$C_{2\varepsilon}^* = C_{2\varepsilon} + \frac{C_\mu \eta^3 (1 - \eta/\eta_0)}{1 + \beta \eta^3}, \quad \eta = \frac{S k}{\varepsilon} \quad (8)$$

where  $S = \sqrt{2 S_{ij} S_{ij}}$  is the mean strain-rate invariant,  $\mu_{eff} = \mu + \mu_t$ , and  $\mu_t = \rho C_\mu k^2 / \varepsilon$ . The RNG variant introduces the additional strain-dependent term in Eq. (8), improving predictions of the high-swirl, rapidly compressing/expanding flow fields characteristic of the sector-mesh in-cylinder domain relative to the standard  $k$ - $\varepsilon$  model.

#### C. Spray Modeling: Lagrangian Droplet Dynamics and KH-RT Breakup

Liquid fuel is tracked using a discrete-droplet Lagrangian formulation coupled to the Eulerian gas phase. Neglecting buoyancy relative to aerodynamic drag, the droplet momentum equation is

$$m_d \frac{d\mathbf{v}_d}{dt} = \frac{1}{2} C_D \rho_g A_d |\mathbf{u}_g - \mathbf{v}_d| (\mathbf{u}_g - \mathbf{v}_d) \quad (9)$$

with the drag coefficient given by the standard sphere-drag correlation,

$$C_D = \begin{cases} \frac{24}{Re_d} \left(1 + \frac{1}{6} Re_d^{2/3}\right), & Re_d \leq 1000 \\ 0.424, & Re_d > 1000 \end{cases} \quad (10)$$

Primary and secondary breakup are governed by the Kelvin–Helmholtz/Rayleigh–Taylor (KH-RT) hybrid model [18], [19]. The fastest-growing KH surface wave has wavelength  $\Lambda_{KH}$  and growth rate  $\Omega_{KH}$ :

$$\frac{\Lambda_{KH}}{a} = 9.02 \frac{(1 + 0.45Z^{0.5})(1 + 0.4T^{0.7})}{(1 + 0.87We_g^{1.67})^{0.6}} \quad (11)$$

$$\Omega_{KH} \left[ \frac{\rho_l a^3}{\sigma} \right]^{0.5} = \frac{0.34 + 0.38We_g^{1.5}}{(1 + Z)(1 + 1.4T^{0.6})} \quad (12)$$

where  $Z = We_l^{0.5}/Re_l$  is the Ohnesorge number,  $T = ZWe_g^{0.5}$  is the Taylor number, and  $We_g$ ,  $We_l$ ,  $Re_l$  are the gas-Weber, liquid-Weber, and liquid-Reynolds numbers based on parent-droplet radius  $a$ . Child droplets of radius  $r_c = B_0\Lambda_{KH}$  are stripped from the parent droplet at rate

$$\frac{da}{dt} = -\frac{a - r_c}{\tau_{KH}}, \quad \tau_{KH} = \frac{3.726 B_1 a}{\Lambda_{KH} \Omega_{KH}} \quad (13)$$

where  $B_0 = 0.61$  and  $B_1$  is a calibration constant typically in the range 1.73–60, recently shown to require recalibration as a function of ambient density and fuel type for accurate biodiesel spray predictions [19]. Rayleigh–Taylor instabilities are tracked concurrently on the decelerating droplet surface; when the RT wave growth time falls below a critical residence time, catastrophic breakup is imposed, generating child droplets whose radius is set by the fastest-growing RT wavelength. This hybrid formulation is reported to give the most physically consistent liquid-penetration and droplet-size predictions among competing breakup models for high-pressure, dense diesel and biodiesel sprays [20].

#### D. Chemical Kinetics: Multi-Fuel Surrogate and Detailed-Chemistry Algorithm

Each elementary reaction  $j$  proceeds at a rate governed by the modified Arrhenius expression

$$k_j = A_j T^{n_j} \exp\left(-\frac{E_j}{R_u T}\right) \quad (14)$$

and the net production rate of species  $k$  used in Eq. (5) is

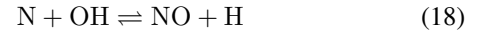
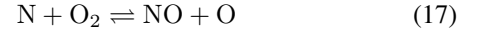
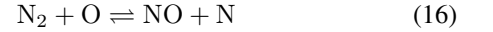
$$\dot{\omega}_k = M_k \sum_{j=1}^{N_r} (\nu''_{kj} - \nu'_{kj}) k_j \prod_{m=1}^{N_{sp}} [X_m]^{\nu'_{mj}} \quad (15)$$

The present work employs a custom multi-fuel mechanism for a biodiesel ternary surrogate of n-heptane, methyl palmitate, and methyl oleate, following the modeling lineage established by Brakora *et al.*, whose two-component methyl-butanoate/n-heptane mechanism (302 g/mol, matching soy-biodiesel’s 292 g/mol molecular weight to within 3%) was the first reduced kinetic scheme validated against neat-biodiesel engine data [24], [25], and subsequently extended to tri-component methyl-decanoate/methyl-9-decenoate/n-heptane skeletal mechanisms

(115 species, 460 reactions) that better represent the degree of unsaturation of real fatty-acid-methyl-ester blends [26]. Because direct integration of  $N_{sp} \sim 100$ -species stiff-ODE chemistry at every computational cell, every time step, is prohibitively expensive in a 3-D sector-mesh domain, detailed chemistry solvers of the SAGE class couple the Eulerian flow solver to the kinetics solver on a per-cell basis while employing a multi-zone acceleration strategy: cells with similar temperature and composition state are dynamically clustered into representative “zones,” the stiff-ODE integration is performed once per zone rather than once per cell, and the result is redistributed to the constituent cells [16], [17]. This reduces the chemistry evaluation cost by one to two orders of magnitude while retaining near-identical accuracy to cell-by-cell integration, and is the algorithmic basis by which ANSYS Forte’s embedded Chemkin-Pro solver achieves tractable run times for the 60°-sector, multi-fuel simulations reported here [14], [17].

#### E. Pollutant Formation Kinetics: Extended Zeldovich Mechanism

Thermal NOx formation, the dominant NOx-formation pathway in diesel and biodiesel CI combustion, is governed by the extended Zeldovich mechanism [21], [22]:



Under the partial-equilibrium assumption for atomic nitrogen (justified because [N] is a highly reactive intermediate whose formation and consumption rates are nearly balanced), the net NO formation rate reduces to

$$\frac{d[NO]}{dt} = 2k_{1f}[O][N_2] \left( \frac{1 - ([NO]/[NO]_e)^2}{1 + (k_{1r}[NO]/(k_{2f}[O_2] + k_{3f}[OH]))} \right) \quad (19)$$

where  $[NO]_e$  is the equilibrium NO concentration and  $k_{1f}$ ,  $k_{2f}$ ,  $k_{3f}$ ,  $k_{1r}$  are the forward/reverse Arrhenius rate constants for reactions (16)–(18), each of the form of Eq. (14) [23]. Because reaction (16) has an activation energy corresponding to the  $N_2$  triple-bond dissociation energy, Eq. (19) is strongly super-linear in temperature and becomes significant only above approximately 2000 K [21] – consistent with the sharp  $NO_2$  onset observed in the B100 simulation beginning at 50° CA ATDC (Section VI), where sustained oxygen-enriched combustion drives local temperatures into this super-linear regime.

## IV. METHODOLOGY

### A. Computational Domain and Geometric Discretization

To optimize computational efficiency while preserving high-fidelity aerodynamic and thermodynamic resolution, the engine geometry was modeled using a 60° sector mesh (Fig. 1) [14]. This approach leverages the inherent rotational symmetry of the multi-hole injector and cylindrical combustion chamber, an approach independently validated against full-geometry and

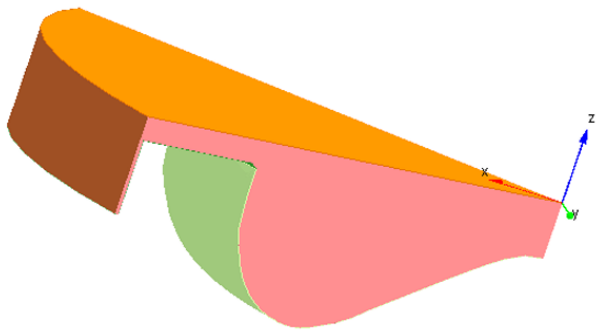


Fig. 1. Computational representation of the 60° sector mesh geometry, highlighting the symmetric boundaries and piston bowl architecture.

experimental datasets in comparable FORTE-based biodiesel studies [12]. The engine architecture is defined by a stroke of 16.51 cm, a connecting rod length of 26.16 cm, and a zero piston offset; the bore diameter and geometric compression ratio were fixed internally within the Sector Mesh Generator consistent with these stroke and rod-length parameters and are not restated here for brevity. The computational domain was partitioned into seven distinct boundary regions: periodic-A, periodic-B, head, liner, crevice, piston-bowl, and piston-crown.

A multifaceted discretization strategy was applied via the ANSYS Sector Mesh Generator to accurately capture boundary layer interactions and steep combustion gradients (Fig. 2). Global point and cone refinements were uniformly set to 1 mm cells. Surface refinements were differentiated by region: the piston, head, and liner surfaces received a single layer of 1 mm cells; the crevice region was resolved with a single layer of 0.25 mm cells; and the squish region incorporated two layers of 0.5 mm cells. A line refinement along the sector mesh axis enforced 0.25 mm cells over a cylindrical radius of 1 mm. Furthermore, Solution Adaptive Meshing (SAM) was activated, utilizing 0.5 mm cells to dynamically resolve critical gradients. SAM targeted the fuel vapor mass fraction and velocity fields from the start of injection ( $-67^\circ$  CA ATDC) to slightly past expected ignition ( $+10^\circ$  CA ATDC), while temperature gradients were adaptively resolved from  $-67^\circ$  CA to  $50^\circ$  CA ATDC.

### B. Initial and Boundary Conditions

The closed-cycle simulation commenced with an initial in-cylinder bulk temperature of 391 K and an absolute pressure of 3.34 bar. The initial fluid dynamics were characterized by an engine swirl ratio of  $-0.7$  and a swirl profile factor of 3.11. The initial gas composition was defined by specific species mole fractions, predominantly  $N_2$  (0.765) and  $O_2$  (0.125), alongside trace constituents (Fig. 3).

Thermal boundary conditions were assigned to emulate steady-state conjugate heat transfer. The cylinder head and piston walls were maintained at 500 K, with piston wall motion actively simulated. The liner geometry boundary was set to a constant 430 K. Symmetric periodic boundary conditions were enforced on the 60° sector faces.

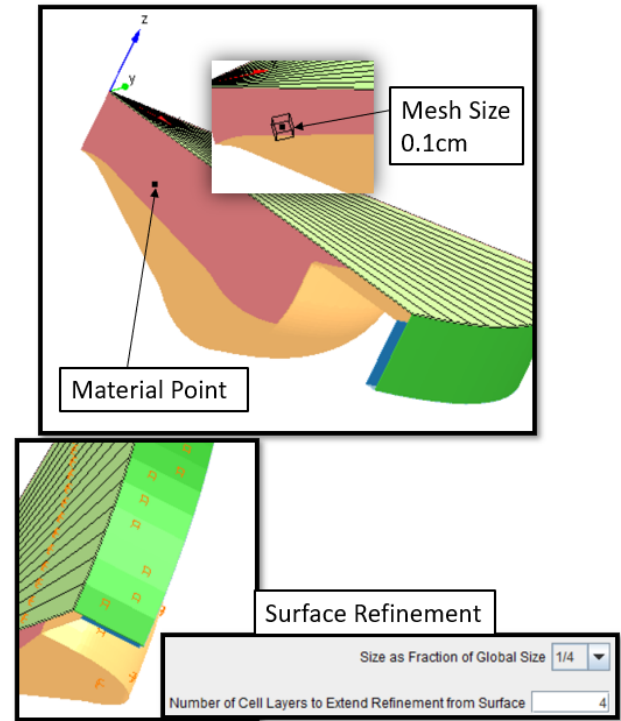


Fig. 2. Discretized mesh illustrating surface refinement protocols and dynamic cell clustering applied to capture critical combustion gradients.

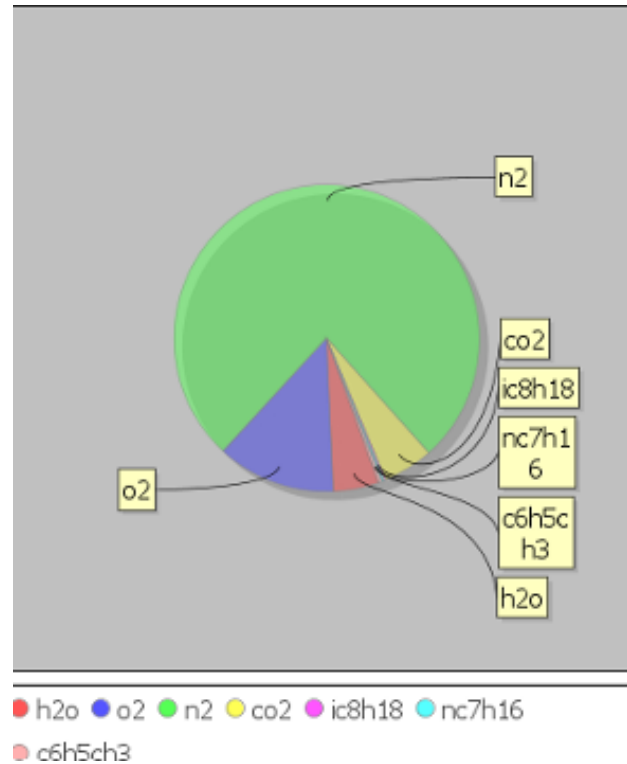


Fig. 3. Initial in-cylinder gas mixture species composition utilized for the computational baseline.

### C. Spray Modeling and Injection Strategy

Fuel atomization, droplet breakup, and vaporization were modeled using the Lagrangian discrete-droplet, KH-RT formulation of Section III with a solid-cone gas-jet injector representation. The nozzle orifice area was parameterized at

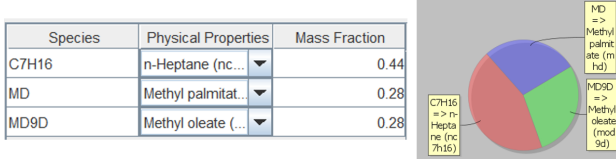


Fig. 4. Mass fraction breakdown of the B100 biodiesel ternary surrogate, encompassing n-heptane, methyl palmitate, and methyl oleate.

0.000491 cm<sup>2</sup>, with a droplet inflow temperature specified at 360 K. A split-injection strategy was employed to facilitate the pre-mixed dual-fuel combustion profile, consistent with common-rail pilot/main strategies reported elsewhere in the biodiesel CFD literature [25]:

- **Pilot Injection (Injection 1):** Actuated from  $-67^\circ$  CA to  $5.46^\circ$  CA, delivering a primary fuel mass of 0.015042 g.
- **Main Injection (Injection 2):** Commencing immediately upon closure of the pilot injection and configured with a duration of  $2.73^\circ$  CA, introducing an additional 0.007521 g of fuel.

#### D. Combustion Modeling and Chemical Kinetics

Combustion kinetics were resolved utilizing the custom multi-fuel reaction mechanism of Section III-D, integrated into the ANSYS chemistry solver. Pure biodiesel (B100) was defined as a ternary surrogate mixture consisting of 44% n-heptane (C<sub>7</sub>H<sub>16</sub>), 28% methyl palmitate, and 28% methyl oleate by mass (Fig. 4), broadly consistent with the ester-fraction ranges used in tri-component biodiesel surrogate mechanisms [26]. Partial blends were linearly scaled: the B50 blend comprised 72.0% C<sub>7</sub>H<sub>16</sub>, 14.0% methyl palmitate, and 14.0% methyl oleate; the B20 blend comprised 88.8% C<sub>7</sub>H<sub>16</sub> with 5.6% of each respective ester.

The kinetic solver utilized an auto-ignition induced flame propagation model. Chemistry activation was triggered subsequent to the start of fuel injection at a threshold temperature of 600 K. Flame propagation was governed by a critical temperature of 2000 K and a critical size coefficient of unity – the same 2000 K threshold above which the extended Zel-dovich mechanism (Eq. (19)) becomes kinetically significant, directly linking the flame-propagation and NO<sub>x</sub> sub-models. The numerical solution advanced at a constant maximum time step of  $5.0 \times 10^{-6}$  s at an engine speed of 1300 RPM.

### V. BLEND OPTIMIZATION FRAMEWORK: LINEAR PROGRAMMING AND MULTI-OBJECTIVE ALGORITHMS

Beyond the forward CFD simulation of fixed B20/B50/B100 blend ratios, the fuel-blend selection problem itself can be posed as a formal optimization problem, following the LP and evolutionary-algorithm frameworks established in the blending literature [27]–[30].

#### A. Linear Programming Formulation for Constrained Blend Selection

Let  $x = [x_D, x_B, x_A]^T$  denote the volumetric fractions of diesel, biodiesel, and an alcohol oxygenate (e.g., butanol) in

the blend, subject to  $x_D + x_B + x_A = 1$ . Following the LP-blending model of Hoang *et al.* [27], an economically optimal blend satisfying fuel-quality standards (e.g., ASTM D975/EN 590) is obtained by solving

$$\min_x c^T x = c_D x_D + c_B x_B + c_A x_A \quad (20)$$

subject to the linear property-blending constraints

$$CN_{\min} \leq CN_D x_D + CN_B x_B + CN_A x_A \leq CN_{\max} \quad (21)$$

$$O_{2,\min} \leq O_{2,D} x_D + O_{2,B} x_B + O_{2,A} x_A \quad (22)$$

$$\nu_{\min} \leq \nu_D x_D + \nu_B x_B + \nu_A x_A \leq \nu_{\max} \quad (23)$$

$$S_D x_D + S_B x_B + S_A x_A \leq S_{\max} \quad (24)$$

$$x_D + x_B + x_A = 1, \quad x_D, x_B, x_A \geq 0 \quad (25)$$

where  $CN$ ,  $O_2$ ,  $\nu$ , and  $S$  denote the cetane number, oxygen mass fraction, kinematic viscosity, and sulfur content of each component, respectively, and  $c$  is the per-liter cost vector. Because Eqs. (20)–(25) are all linear in  $x$ , the feasible region is a convex polytope and the global optimum lies at a vertex, solvable via the simplex method or an interior-point algorithm. This LP formulation, augmented with an Analytical Hierarchy Process (AHP) weighting of qualitative criteria, was shown to identify a 70% diesel/20% biodiesel/10% butanol blend as jointly cost-, performance-, and emission-optimal, achieving 35% lower sulphur content and 36% lower CO<sub>2</sub> emissions relative to neat diesel [27]. For the present B20/B50/B100 dataset, Eqs. (21)–(23) provide a mathematically rigorous basis for interpolating the CFD-predicted CO, UHC, and NO<sub>2</sub> trends of Section VI onto intermediate blend ratios subject to fuel-quality constraints, rather than relying on the linear mass-fraction scaling used for the surrogate composition alone.

#### B. Multi-Objective Genetic Algorithm Formulation

The LP model of Eqs. (20)–(25) captures only linear blending constraints and a single scalar cost objective; it cannot directly capture the nonlinear, competing trade-off between UHC/CO suppression and thermal-NO<sub>x</sub> generation identified in Section VII. This motivates a multi-objective formulation

$$\min_{x \in \mathcal{X}} \mathbf{F}(x) = [-\text{BTE}(x), \text{BSFC}(x), \text{NO}_x(x), \text{CO}(x), \text{HC}(x)]^T \quad (26)$$

where  $\mathcal{X}$  is the feasible blend-ratio/operating-parameter space (e.g., biodiesel fraction, nano-additive dosage, injection timing) and each objective is a nonlinear, generally non-convex function of  $x$  obtained either from CFD (as in Section VI) or from a regression surrogate (RSM/ANN) fitted to CFD or experimental data, as employed in comparable optimization studies [29], [30]. Problem (26) is solved using a Pareto-dominance genetic algorithm (e.g., NSGA-II/NSGA-III), in which a population of candidate blends  $\{x_k\}_{k=1}^{N_{pop}}$  is evolved via:

- 1) *Non-dominated sorting*: partitioning the population into Pareto fronts  $\mathcal{F}_1, \mathcal{F}_2, \dots$  such that  $x_i$  dominates  $x_j$

(written  $x_i \prec x_j$ ) iff  $F_m(x_i) \leq F_m(x_j) \forall m$  with strict inequality for at least one objective;

- 2) *Crowding-distance assignment*, promoting diversity along each front:

$$d(x_i) = \sum_{m=1}^M \frac{F_m(x_i^{+1}) - F_m(x_i^{-1})}{F_m^{\max} - F_m^{\min}} \quad (27)$$

- 3) *Genetic operators*: simulated-binary crossover and polynomial mutation generate offspring, which are merged with the parent population and re-sorted by (front, crowding-distance) rank to select the next generation.

Applying this scheme to biofuel-blend/operating-parameter optimization, NSGA-III and AGE-MOEA algorithms have identified Pareto-optimal solutions favoring 20–40% biokerosene-biodiesel blends across 50–75% engine load [28], while an ANN-RSM surrogate combined with ACKTR-DE and HHO metaheuristics identified a 100% biodiesel ratio at 180 bar injection pressure as jointly minimizing specific fuel consumption and exhaust emissions for waste-cooking-oil biodiesel [30]. Applied to the present B20/B50/B100/nano-additive dataset, this framework provides the mathematical machinery by which the UHC/CO-versus-NO<sub>x</sub> trade-off identified in Fig. 9 and Fig. 8 can be formally resolved into a Pareto front, rather than the single-blend qualitative recommendation offered in Section VIII.

## VI. RESULTS

### A. In-Cylinder Thermodynamic Characteristics

The simulated in-cylinder pressure and temperature traces indicate that biodiesel blends closely mirror the thermodynamic performance of pure diesel, validating their utility as direct substitute fuels, consistent with the near-identical peak-pressure trends reported for FORTE-simulated waste-oil biodiesel blends [12]. Peak cylinder pressures (Fig. 5) universally approximated 8 MPa near Top Dead Center (TDC). The B100 blend exhibited a marginally higher peak pressure and a steeper rate of pressure rise compared to pure diesel, correlating with the oxygenated nature of the fuel promoting accelerated early-stage combustion.

Similarly, in-cylinder temperature profiles (Fig. 6) demonstrated rapid localized heating upon ignition, with peak macroscopic temperatures reaching approximately 2050 K – above the  $\sim 2000$  K threshold at which the extended Zeldovich mechanism (Eq. (19)) becomes kinetically active. During the expansion stroke, the B100 and B20 blends maintained higher sustained bulk temperatures relative to pure diesel, indicating prolonged late-stage oxidation of the ester components.

### B. Emission Characteristics

The comparative emission analysis reveals profound trade-offs inherent to the oxygen content of the biodiesel surrogate blends, consistent with the oxygenate-driven CO/HC-versus-NO<sub>x</sub> trade-off documented broadly across the nano-additive and alcohol-blend literature (Table I) [4], [5].

**Carbon Monoxide (CO) and Unburnt Hydrocarbons (UHC):** Biodiesel blends demonstrated a significant advantage

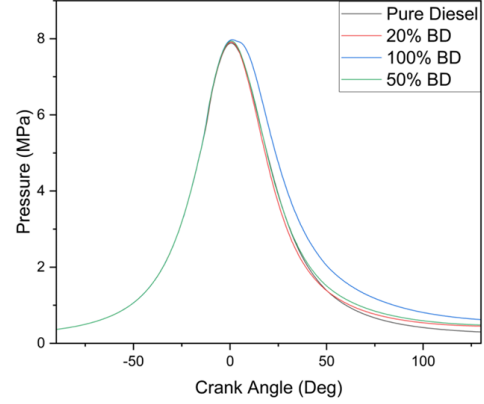


Fig. 5. In-cylinder pressure traces against crank angle, demonstrating comparable peak pressures across all tested fuel blends near TDC.

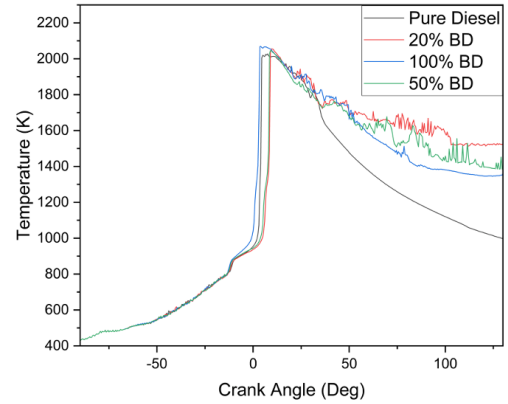


Fig. 6. Macroscopic in-cylinder temperature profiles. Biodiesel blends maintain sustained higher temperatures during the expansion stroke.

in mitigating incomplete combustion products. While intermediate CO formation peaked near 280 g/kg fuel for the B50 blend during early expansion, the B100 blend exhibited rapid late-cycle oxidation (Fig. 7). By the exhaust valve opening phase, B100 achieved a substantially lower terminal CO concentration compared to pure diesel – qualitatively matching the 16–30% experimental CO reductions reported for oxygenate- and nano-additive-doped biodiesel blends [1], [4], [6].

Correspondingly, UHC emissions for B100 plummeted to roughly 0.02 g/kg fuel late in the cycle (Fig. 8), whereas pure diesel plateaued at a significantly higher 0.08 g/kg fuel – a 75% relative reduction, exceeding the 20–28% HC reductions typically reported for nano-additive-doped blends [1], [4], [5] but consistent in direction with the oxygenate-driven UHC suppression mechanism common to both.

**Nitrogen Dioxide (NO<sub>2</sub>):** The primary adverse effect of the oxygenated biodiesel blends was manifested in NO<sub>2</sub> formation (Fig. 9). The B100 blend generated a severe spike in NO<sub>2</sub> emissions (exceeding 0.011 g/kg fuel) beginning at 50° CA ATDC, coincident with the point at which sustained bulk temperature (Fig. 6) crosses the  $\sim 2000$  K Zeldovich activation threshold. In contrast, pure diesel and lower-concentration

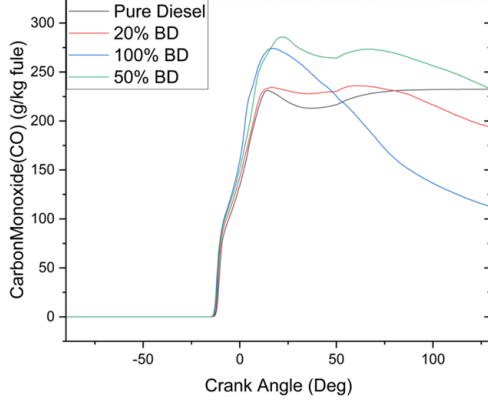


Fig. 7. Carbon Monoxide (CO) formation over the combustion cycle. B100 achieves the lowest terminal CO output due to efficient late-stage oxidation.

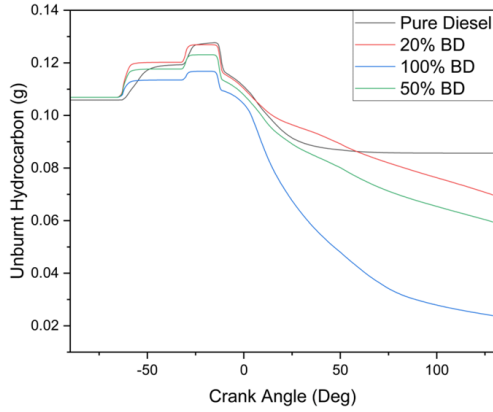


Fig. 8. Unburnt Hydrocarbon (UHC) emissions, highlighting the superior oxidation completion of the B100 blend relative to pure diesel.

blends (B20) suppressed  $\text{NO}_2$  formation to below 0.004 g/kg fuel. This directionally matches the 7–9%  $\text{NO}_x$  increases reported for nano-additive/oxygenate combinations that similarly raise peak combustion temperature and available oxygen [4], while the  $\text{NO}_x$ -suppressing counter-examples reported for CNT and  $\text{Al}_2\text{O}_3$  doping (Table I) point toward nano-additive dosing as a candidate mitigation strategy, addressed further in Section VII.

## VII. DISCUSSION

The computational fluid dynamics results indicate that the integration of oxygenated biodiesel blends fundamentally alters the in-cylinder thermodynamic trajectory during the expansion stroke. As observed in the pressure traces (Fig. 5), the B100 blend achieves a marginally higher peak cylinder pressure and maintains a consistently higher pressure profile throughout late-cycle expansion compared to pure diesel. This prolonged pressure retention correlates directly with the temperature profiles (Fig. 6), wherein B100 and B50 blends sustain bulk in-cylinder temperatures significantly higher than pure diesel during the expansion phase (maintaining approximately 1400 K late in the cycle, whereas pure diesel drops

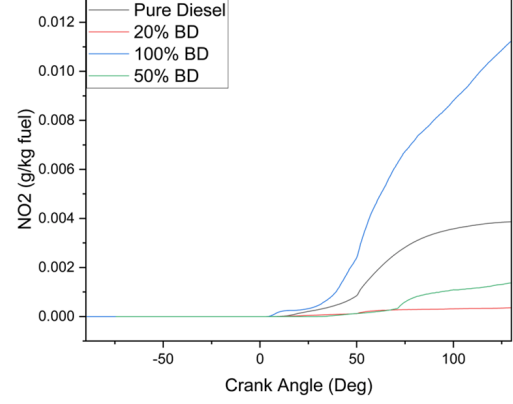


Fig. 9. Nitrogen Dioxide ( $\text{NO}_2$ ) emissions. The high oxygen concentration and sustained temperatures in B100 trigger a substantial thermal  $\text{NO}_x$  penalty.

toward 1000 K). This thermodynamic behavior suggests that the fuel-bound oxygen inherent to the methyl palmitate and methyl oleate components facilitates a more complete, albeit slightly delayed, late-stage oxidation of the fuel charge, consistent with the mechanistic explanation offered for oxygenate-driven combustion enhancement in the nano-additive literature [1], [3].

The emission profiles reveal a distinct trade-off governed by the oxygen content of the surrogate blends, highlighting the classic dynamic between carbonaceous oxidation and thermal nitrogen-oxide generation formalized by Eq. (19). The most pronounced benefit of higher biodiesel concentrations is the suppression of UHC and terminal CO. The UHC mass fraction for B100 plummets continuously during expansion (Fig. 8), terminating at approximately 0.02 g/kg fuel, representing a drastic reduction compared to the pure diesel baseline. While intermediate CO formation during the primary premixed combustion phase peaks heavily for all blends, the late-cycle oxidation kinetics highly favor the B100 blend (Fig. 7). The sustained elevated temperatures combined with surplus oxygen enable B100 to rapidly oxidize CO into  $\text{CO}_2$ , in direct agreement with the CO-reduction mechanism (via enhanced oxygen availability and atomization surface area) reported for both oxygenated alcohol blends and metal-oxide nano-additives [4], [8].

Conversely, these exact mechanisms – elevated sustained temperatures and increased oxygen availability – precipitate a severe penalty in  $\text{NO}_2$  emissions via the extended Zeldovich mechanism (Eqs. (16)–(19)). The B100 blend exhibits a steep, unmitigated accumulation of  $\text{NO}_2$  (Fig. 9) beginning at 50° CA ATDC. Interestingly, the B20 and B50 blends effectively suppress  $\text{NO}_2$  formation, with B20 producing near-zero  $\text{NO}_2$ . This non-linear response suggests an optimal blending threshold where sufficient oxygen is present to reduce UHC and CO without crossing the critical temperature-duration threshold that heavily triggers thermal  $\text{NO}_2$  formation – precisely the trade-off that the multi-objective formulation of Eq. (26) is designed to resolve quantitatively rather than qualitatively. Notably, the literature-reported ability of  $\text{Al}_2\text{O}_3$  and CNT nano-additives to *reduce*  $\text{NO}_x$  by up to 3.92%, even while

raising or maintaining BTE [2], [3], suggests that co-doping the B100 blend with a modest concentration ( $\sim 50$ – $150$  ppm, per Table I) of an NO<sub>x</sub>-suppressing nanoparticle could decouple the UHC/CO benefit of high biodiesel fraction from its NO<sub>x</sub> penalty – a hypothesis directly testable within the CFD framework of Section III by adding a nanoparticle-conditioned Arrhenius pre-exponential correction to Eq. (14), as has been proposed in the nano-additive kinetics literature [4].

### VIII. CONCLUSION

This computational investigation utilized ANSYS Forte to rigorously evaluate the combustion dynamics and emission profiles of a compression ignition engine operating on varying concentrations of a ternary biodiesel surrogate, and situates these results within the governing conservation equations, RNG  $k$ - $\epsilon$  turbulence closure, KH-RT spray breakup kinetics, detailed multi-fuel chemistry, and extended-Zeldovich NO<sub>x</sub> formulation that collectively underpin the CFD solver. The implementation of a  $60^\circ$  sector mesh with dynamic cell clustering successfully captured the complex interactions of auto-ignition, flame propagation, and species formation without prohibitive computational expense, consistent with comparable sector-mesh biodiesel CFD studies in the literature [12], [14].

The findings establish that biodiesel blends, particularly at higher concentrations (B50 and B100), serve as highly capable thermodynamic substitutes for conventional diesel, maintaining optimal peak pressures while sustaining elevated temperatures during the expansion stroke. However, the viability of these fuels is strictly dictated by the oxygen-emission trade-off quantified by the extended Zeldovich kinetics of Section III. While B100 effectively neutralizes unburnt hydrocarbons (75% reduction relative to diesel) and mitigates terminal carbon monoxide through enhanced late-cycle oxidation, it incurs a severe NO<sub>2</sub> penalty driven by sustained high-temperature thermal mechanisms. The data strongly indicates that intermediate blends, specifically B20 and B50, offer an optimal stoichiometric compromise, a conclusion reinforced by the Linear-Programming and multi-objective genetic-algorithm formulations of Section V, which provide a rigorous mathematical pathway – rather than a purely qualitative one – for identifying blend ratios and nano-additive dosages that Pareto-optimally balance BTE, BSFC, CO, HC, and NO<sub>x</sub>. Future experimental and computational efforts must focus on coupling B100 utilization with advanced exhaust gas recirculation (EGR) strategies, targeted NO<sub>x</sub>-suppressing nano-additive doping (e.g., Al<sub>2</sub>O<sub>3</sub> or CNT at 50–150 ppm, per Table I), and explicit NSGA-II/III-based Pareto optimization of blend ratio and injection timing to quench late-cycle temperatures and mitigate the nitrogen-oxide penalty while preserving the carbonaceous-emission benefits demonstrated here.

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